

# MOHAMMED SOHAIL RASUL

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## EDUCATION

**Manipal Institute of Technology, Manipal**

*BTech in Mathematics and Computing*

2024 – 2028

*CGPA: 8.31*

## EXPERIENCE

**Arogyam GridTech Private Limited**

*AI Developer Intern*

Apr 2026 – Present

*Remote*

- Contributing to ground-up development of an AI voice agent; evaluating and integrating third-party APIs using Python
- Collaborating with partner companies to assess and select the optimal API stack for the voice pipeline
- Transitioning to LLM/AI pipeline layer – responsible for prompt engineering and model integration after core infrastructure is established

## PROJECTS

**Stock Price Analysis & Backtesting** | *Python, Pandas, NumPy, Matplotlib* | Github

2026

- Built a backtesting engine using **NumPy** to simulate SMA crossover strategies on historical OHLCV data.
- Computed key risk-adjusted metrics — **Sharpe Ratio**, **Maximum Drawdown**, and **Annualized Volatility** — benchmarked against a buy-and-hold baseline.
- Visualized equity curves and signal overlays to communicate strategy performance and risk-reward trade-offs clearly.

**Credit Risk Classifier** | *Python, Scikit-learn, XGBoost, Pandas* | Github

2026

- Trained a default probability classifier on a **150K borrower dataset**, achieving **0.868 ROC-AUC** using XGBoost.
- Tackled a **93:7 class imbalance** via cost-sensitive learning and threshold tuning to improve recall on high-risk borrowers.
- Performed feature engineering and data auditing to handle missing financial disclosures and reduce outlier impact on model stability.

**E-commerce & BMW Sales Analysis** | *Python, Pandas, PowerBI, Seaborn* | Github

2025

- Executed end-to-end exploratory analysis on diversified datasets, identifying key revenue drivers and consumer purchasing patterns in the luxury automotive and retail sectors.
- Correlated pricing dynamics with sales volume to optimize inventory strategies and regional market penetration.
- Synthesized raw data into **PowerBI dashboards**, translating complex multivariate trends into visual narratives for data-driven stakeholder decision-making.

## TECHNICAL SKILLS

**Languages:** Python, C, SQL

**Libraries:** Pandas, NumPy, Scikit-learn, Matplotlib

**Tools:** Jupyter, Git, PowerBI

**Concepts:** Machine learning, backtesting, EDA, regression, Data Analysis

## ACHIEVEMENTS

**1st Place, Generative AI** | M#, Manipal Hackathon 2025

## RELEVANT COURSEWORK

Data Structures & Algorithms · Operating Systems · Probability & Statistics · Linear Algebra · Database Systems ·